

MINGEUN CHOI

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EDUCATION

Georgia Institute of Technology

Ph.D. in Mechanical Engineering (Advisor: Dr. Satish Kumar)

Atlanta, GA
Jan. 2023 – Dec. 2027 (expected)

Seoul National University

M.S. in Mechanical Engineering (Advisor: Dr. Seung Jin Song)

Seoul, Republic of Korea

Korea University

B.S. in Mechanical Engineering

Mar. 2017 – Feb. 2019

Seoul, Republic of Korea

Mar. 2013 – Feb. 2017

RESEARCH EXPERIENCE (SELECTED)

Graduate Research Assistant, Georgia Institute of Technology

Jan. 2023 – Present

Collaborating with an interdisciplinary team of material scientists and electrical engineers to explore a self-consistent multiscale workflow bridging ballistic-to-continuum physics and device-to-package engineering, achieving the following:

- Designed an automated electrothermal modeling framework to co-optimize vertical power delivery (VPD) architectures and substrate-embedded microfluidic cooling for thermally sustainable, high-power-density operations.
- Developed deep operator network (DeepONet)-based surrogate models for FinFET arrays and VPD packages to rapidly predict spatiotemporal temperature profiles for thermally-aware early design exploration.
- Developing a physics-informed neural network (PINN)-based surrogate model for frequency-domain thermorefectance (FDTR) to estimate depth-dependent thermal properties of thin films for high-throughput, accurate characterizations.

TECHNICAL SKILLS

- **Engineering and Simulation Software:** Ansys Electronics Desktop (AEDT) Icepak (including PyAEDT); Ansys Fluent (including PyFluent); Dassault Systèmes SolidWorks; Microsoft Visual Studio Code.
- **Programming and Scripting:** Python; MathWorks MATLAB; Julia; National Instruments LabVIEW.
- **Experimental Metrology:** Frequency-domain thermorefectance (FDTR); thermorefectance thermal imaging (TTI); thermofluidic measurement; machining; manufacturing; commissioning; calibration; uncertainty analysis.

HONORS AND AWARDS (SELECTED)

Perspective or Roadmap Paper Topping Fellowship , Georgia Institute of Technology	2026
News Release Most First Author Journal Papers Topping Fellowship, Georgia Institute of Technology	2025
Best Paper Runner-Up Award , Institute of Electrical and Electronics Engineers	2025
News Release Excellence in Service and Leadership Award, Georgia Institute of Technology	2025
Community Builder/Leader Topping Fellowship , Georgia Institute of Technology	2025
Strong Gap Funding Topping Fellowship , Georgia Institute of Technology	2024
Best Presentation Award , Korean Society for Fluid Machinery	2018
Merit-Based Scholarship , Seoul National University	2018
Brain Korea 21 Plus Fellowship , National Research Foundation of Korea	2017
National Science and Engineering Scholarship , Korea Student Aid Foundation	2015 – 2016
Academic Excellence Scholarship for Freshman , Korea University	2013

LEADERSHIP AND SERVICE (SELECTED)

Director of Communications, [Georgia Tech Nu Chapter of Pi Tau Sigma](#)

Jan. 2025 - Dec. 2025

- Launched a monthly newsletter and renovated the chapter website, driving a 100 % increase in new-initiate turnout in Spring 2025 and boosting alumni engagement by 30 %.

External Vice President, [Mechanical Engineering Graduate Association \(MEGA\)](#)

May 2024 - Apr. 2025

- Coordinated eight professional networking events with Dow Inc., Sandia National Laboratories (SNL), Exponent, and Kratos Defense, increasing event offerings by 50 % and graduate-student attendance by 40 % over the academic year.

Refereed Journal Articles

- [8] **M. Choi**, S. Krishnakumar, Y. Popryho, R. R. Khorasani, M. Swaminathan, I. Partin-Vaisband, and S. Kumar, "Self-Consistent Electrothermal Modeling of Distributed Vertical Power Delivery Architecture with Substrate-Embedded Microfluidic Cooling," *IEEE Trans. Compon., Packag., Manuf. Technol.*, early access, Nov. 24, 2025.
- [7] D. Go, V. Ashita, A. Wang, **M. Choi**, H. S. Song, S. Yun, D. Pal, P.-C. Lee, J. Dutta, J. Ryu, S. Kumar, C. V. Thompson, M. S. Bakir, A. C. Kummel, "Chiplet Encapsulation and Planarization with 100 μm Deep Gaps using AlN Powder Spray + Sputter Hybrid Deposition," *Adv. Eng. Mater.*, vol. 28, no. 6, pp. 1438-1656, Mar. 2026.
- [6] S. Yun, D. Go, **M. Choi**, R. Kondakindi, P.-C. Lee, P. R. Bandaru, S. Kumar, and A. C. Kummel, "High Speed, High Thermal-Conductivity of Aluminum Nitride Deposited by DC Reactive Sputtering at Low Temperature in the Transition Regime," *Ceram. Int.*, vol. 52, no. 4, pp. 4777-4786, Feb. 2026.
- [5] P.-C. Lee, **M. Choi**, D. Contreras Mora, K. Wang, S. Yun, D. Go, J. Dutta, D. Pal, S. Kumar, and A. C. Kummel, "High-Speed AlN Film Deposition via Low-Pressure Bipolar High Power Impulsed Magnetron Sputter for Enhanced Thermal Conductivity," *Thin Solid Films*, vol. 832, p. 140821, Dec. 2025.
- [4] **M. Choi**, D. Vaca, and S. Kumar, "Machine Learning-Enabled Fast and Accurate Inversion of Thermal Properties at the Micro- and Nanoscale via Optical Metrology," *Annu. Rev. Heat Transfer*, vol. 28, pp. 277-337, Oct. 2025.
- [3] **M. Choi**, S. Krishnakumar, R. R. Khorasani, M. Swaminathan, I. Partin-Vaisband, and S. Kumar, "Substrate-Embedded Microfluidic Cooling of Distributed Vertical Power Delivery Architectures for High-Performance Computing Processors," *IEEE Trans. Compon., Packag., Manuf. Technol.*, vol. 15, no. 9, pp. 1912-1920, Sep. 2025.
- [2] P.-C. Lee, A. J. McLeod, **M. Choi**, D. Vaca, D. Contreras Mora, K. Wang, S. Yun, J. Dutta, D. Pal, S. Kumar, and A. C. Kummel, "Achieving a High Thermally Conductive One Micron AlN Deposition by High Power Impulse Magnetron Sputtering Plus Kick," *ACS Appl. Mater. Interfaces*, vol. 16, no. 20, pp. 26664-26673, May 2024.
- [1] **M. Choi**, B. Goo, G. Cho, and S. J. Song, "Swirl Enhancement Effect on Turbine Rim Seal Performance," *ASME J. Turbomach.*, vol. 146, no. 5, p. 051002, May 2024.

Refereed Conference Proceedings

- [7] **Best Paper** **M. Choi**, M. Herndon, K. Uechi, N. Akel, and S. Kumar, "DeepONet-Enabled Fast and Scenario-Aware Spatial Temperature Prediction of a Vertical Power Delivery Module for Hyperscale Data Centers," in *Proc. 25th IEEE Intersoc. Conf. Thermal Thermomech. Phenomena Electron. Syst. (ITherm)*, Orlando, FL, USA, May. 2026, to be published.
- [6] R. Dutta, M. Shon, S. Lim, P. Saha, **M. Choi**, M. P. Singh, S. Kumar, S. Mukhopadhyay, S. K. Lim, S. Yu, and S. Datta, "Self-Heating in GAA Nanosheet FETs with integrated Back Side Power Delivery Network: Device to Circuit Electro-Thermal Analysis and Implications," in *Proc. IEEE Int. Reliab. Phys. Symp. (IRPS)*, Tucson, AZ, USA, Mar. 2026.
- [5] **Best Paper** **News Release** **M. Choi**, S. Krishnakumar, Y. Popryho, R. R. Khorasani, M. Swaminathan, I. Partin-Vaisband, and S. Kumar, "Automated Electro-Thermal Modeling Framework of Distributed Vertical Power Delivery Architectures with Substrate-Embedded Microfluidic Cooling," in *Proc. 24th IEEE Intersoc. Conf. Thermal Thermomech. Phenomena Electron. Syst. (ITherm)*, Dallas, TX, USA, May 2025.
- [4] **News Release** **M. Choi**, R. Dutta, P. Saha, M. P. Singh, S. Mukhopadhyay, S. Datta, and S. Kumar, "Fast Prediction of Spatio-Temporal Temperature Profiles in FinFET Arrays via Numerical and Machine-Learning Approaches," in *Proc. IEEE Int. Electron Devices Meeting (IEDM)*, San Francisco, CA, USA, Dec. 2024.
- [3] Y. Im, J. Kim, **M. Choi**, M. Bouzidi, X. Li, J. W. Kim, A. M. Muslu, S. Kumar, M. Swaminathan, S. K. Sitaraman, and Y. Joshi, "Parametric Thermal Design for Heterogeneously Integrated High-Power Packages," in *Proc. ASME 2024 Int. Tech. Conf. Exhib. Packaging Integr. Electron. Photon. Microsystems (InterPACK)*, San Jose, CA, USA, Oct. 2024.
- [2] S. Krishnakumar, **M. Choi**, R. R. Khorasani, R. Sharma, M. Swaminathan, S. Kumar, and I. Partin-Vaisband, "Vertical Power Delivery for High Performance Computing Systems with Buck-Derived Regulators," in *Proc. IEEE 74th Electron. Compon. Technol. Conf. (ECTC)*, Denver, CO, USA, May 2024.
- [1] **M. Choi**, S. Krishnakumar, R. R. Khorasani, I. Partin-Vaisband, R. Sharma, M. Swaminathan, and S. Kumar, "Thermal Analysis of High Current Vertical Power Delivery Network with Embedded Microchannel Cooling," in *Proc. 23rd IEEE Intersoc. Conf. Thermal Thermomech. Phenomena Electron. Syst. (ITherm)*, Aurora, CO, USA, May 2024.

Patents

- [1] A. C. Kummel, P.-C. Lee, D. Contreras Mora, D. Go, S. Kumar, and **M. Choi**, "Aluminum Nitride Thin Film Deposition Using Sputtering," U.S. Patent Application No. 19/366,019, filed on Oct. 22, 2025, published on Apr. 23, 2026.